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Secure personal information exchange by an information handling system

Abstract

Computer-based information security techniques are provided that enable a user to reduce and control the amount of personal information digitally supplied to any given service provider when transacting or otherwise interacting with these service providers. Using these techniques, the user can release the minimally required personal information needed for a particular service provider to perform a particular service, without a link to a public identity of the user. Stated differently, only the information that a particular service provider needs in order to complete a service provided by the service provider is conveyed to the service provider based on the user's policy settings. An encryption scheme is used to securely transfer the various portions of personal information to the various service providers.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8656037	12/2013	Itoh	380/282	H04L 67/02
9628268	12/2016	Kiang	N/A	G06F 21/6218
10560977	12/2019	Motto	N/A	H04L 63/08
2008/0056501	12/2007	McGough	380/282	H04L 63/08
2014/0064486	12/2013	Abraham	380/270	H04W 12/03
2017/0093842	12/2016	Bent	N/A	H04L 63/0807
2019/0140844	12/2018	Brown	N/A	H04L 63/0428
2021/0035669	12/2020	Culver	N/A	G06F 9/547
2021/0241355	12/2020	Low	N/A	H04L 63/062
2023/0379155	12/2022	Xue	N/A	G06F 21/64

OTHER PUBLICATIONS

NPL Search Terms (Year: 2025). cited by examiner

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The instant disclosure relates to information handling systems. More specifically, portions of this disclosure relate to information security techniques for user data control.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow

for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(3) Users today digitally interact with many service providers, including social media, commerce services, etc. It is often useful or necessary for these service providers to have knowledge of certain personal information of the user in order to provide the user with the service they desire. Typically, however, transmitting personal information to these service providers creates risk to user privacy because the personal information becomes the property of these individual service providers. For example, each service provider that a user creates an account with typically stores in a respective platform database a list of personal information (e.g., given name, address, contact information, etc.) associated with a platform username of the user. Only one of these databases need be compromised in order for that personal information to become public domain, or at least available to an opportunistic or malicious entity, which increases the risk of a privacy breach and identity theft for the user. Typical techniques attempt to protect this personal information of a user through encryption, but conventional network architecture design, physical security policies, and human error have proven that these typical techniques are vulnerable to compromise.

(4) Additionally, insights about user preferences are often not shared between platforms of different service providers. As such, users often have differing tiers of experience between these different service providers. The lack of sharing also results in the practice of these service providers hoarding and selling the personal information of the user, which the user has no technological control over.

SUMMARY

(5) Computer-based techniques are provided that enable a user to reduce the amount of personal information digitally supplied to any given service provider when transacting or otherwise interacting with these service providers. Using these techniques, the user can release the minimally required personal information needed for a particular service provider to perform a particular service, without a link to a public identity of the user. Stated differently, only the information that a particular service provider needs in order to complete a service provided by the service provider is conveyed to the service provider based on the user's policy settings. In an example, a second service provider may perform a second service only after a first service provider completes a first service. In such an example, the user may release to the first service provider only the personal information that the first service provider needs to complete the first service, and then release to the second service provider only the personal information that the second service provider needs to complete the second service. The user could authorize each transmittal of information individually or could set preferences for a pre-approved data flow such that an information management service automatically communicates with various third party service providers based on the user-set preferences. The user may also have the option to opt-out of providing the personally identifying information or utilize a brokering service to maintain complete anonymity relative to a service provider.

(6) In this way, the present computer-based techniques improve information security technology. For example, the techniques provide users with technology-based control over their digitally-stored personal information. Because there is a single source of personal information that the user controls to digitally release certain information to certain service providers, the user experience can be consistent across the various platforms of various service providers. Additionally, because the service providers are provided with the personal information these service providers need in order to provide the service the users expect, users can transact in a way that is personalized yet anonymous, which can return agency of personal data back to individual users. The techniques also reduce the risk of a privacy breach or identity theft for the user by limiting the amount of personal

information that is at risk from any one breach of a service provider's network. For example, if a malicious actor were to steal the user's address from a delivery service provider's network, all the malicious actor would have is an address that is unassociated with anything else except possibly an anonymous identifier. The address would not be associated with the public identity of the user, the user's contact information, the user's payment information, or any other personal information of the user because the delivery service provider's network didn't have that information because that information was not needed to make a delivery to the user's address.

(7) Conventional computer-based information security attempts to safeguard an identity of a user include providing a decentralized identity that allows a user to manage their own identity. Such decentralized identities, however, do not give the user full control over the personally identifying information of the user through a personal signing mechanism of the user.

(8) Shortcomings mentioned here are only representative and are included to highlight problems that the inventors have identified with respect to existing information handling systems and sought to improve upon. Aspects of the information handling systems described below may address some or all of the shortcomings as well as others known in the art. Aspects of the improved information handling systems described below may present other benefits than, and be used in other applications than, those described above.

(9) According to one embodiment, an information handling system comprises a memory storing a private key and personally identifying information associated with a user; and a processor in communication with the memory. The processor is configured to perform operations including receiving a first encrypted request for a first portion of the personally identifying information associated with the user. The encrypted request is encrypted using a first public key of a first key pair including the private key and the first public key. A completion of a first service by a first service provider is based on the first portion. The first service is part of a process that includes the first service performed by the first service provider and a second service performed by a second service provider different than the first service provider. The second service is different than the first service, and the second service is performed by the second service provider based on a completion of the first service by the first service provider. The operations further include decrypting the first encrypted request using the first private key; encrypting the first portion of the personally identifying information using a second public key associated with the first service provider; and transmitting the first portion that is encrypted to a first computing device associated with the first service provider. Subsequent to a completion of the first service by the first service provider, the operations further include receiving a second encrypted request for a second portion of the personally identifying information associated with the user. The second encrypted request is encrypted using the first public key, the second portion is different than the first portion, and a completion of the second service by the second service provider is based on the second portion. The operations further include decrypting the second encrypted request using the first private key; encrypting the second portion of the personally identifying information using a third public key associated with the second service provider; and transmitting the second portion that is encrypted to a second computing device associated with the second service provider.

(10) In certain embodiments, the personally identifying information associated with the user consists of the first portion, the second portion, and a remaining portion. The second portion and the remaining portion are not transmitted to the first computing device.

(11) In certain embodiments, of the personally identifying information associated with the user, the completion of the first service requires only the first portion.

(12) According to another embodiment, a method may include receiving a first encrypted request for a first portion of personally identifying information associated with a user. The encrypted request is encrypted using a first public key of a first key pair including a private key and the first public key. A completion of a first service by a first service provider is based on the first portion. The first service is part of a process that includes the first service performed by the first service

provider and a second service performed by a second service provider different than the first service provider. The second service is different than the first service, and the second service is performed by the second service provider based on a completion of the first service by the first service provider. The method further includes decrypting the first encrypted request using the first private key; encrypting the first portion of the personally identifying information using a second public key associated with the first service provider; and transmitting the first portion that is encrypted to a first computing device associated with the first service provider. Subsequent to a completion of the first service by the first service provider, the method further includes receiving a second encrypted request for a second portion of the personally identifying information associated with the user. The second encrypted request is encrypted using the first public key, the second portion is different than the first portion, and a completion of the second service by the second service provider is based on the second portion. The method further includes decrypting the second encrypted request using the first private key; encrypting the second portion of the personally identifying information using a third public key associated with the second service provider; and transmitting the second portion that is encrypted to a second computing device associated with the second service provider.

(13) The method may be embedded in a computer-readable medium as computer program code comprising instructions that cause a processor to perform operations corresponding to the steps of the method. In some embodiments, the processor may be part of an information handling system including a first network adaptor configured to transmit data over a first network connection; and a processor coupled to the first network adaptor, and the memory.

(14) As used herein, the term “coupled” means connected, although not necessarily directly, and not necessarily mechanically; two items that are “coupled” may be unitary with each other. The terms “a” and “an” are defined as one or more unless this disclosure explicitly requires otherwise. The term “substantially” is defined as largely but not necessarily wholly what is specified (and includes what is specified; e.g., substantially parallel includes parallel), as understood by a person of ordinary skill in the art.

(15) The phrase “and/or” means “and” or “or”. To illustrate, A, B, and/or C includes: A alone, B alone, C alone, a combination of A and B, a combination of A and C, a combination of B and C, or a combination of A, B, and C. In other words, “and/or” operates as an inclusive or.

(16) The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), and “include” (and any form of include, such as “includes” and “including”) are open-ended linking verbs. As a result, an apparatus or system that “comprises,” “has,” or “includes” one or more elements possesses those one or more elements, but is not limited to possessing only those elements. Likewise, a method that “comprises,” “has,” or “includes,” one or more steps possesses those one or more steps, but is not limited to possessing only those one or more steps.

(17) Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the present application, discussions utilizing the terms such as “accessing,” “receiving,” “sending,” “using,” “selecting,” “determining,” “normalizing,” “multiplying,” “averaging,” “monitoring,” “comparing,” “applying,” “updating,” “measuring,” “deriving,” “settling,” “generating” or the like, refer to the actions and processes of a computer system, audio controller, or similar electronic computing device that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system's registers, memories, or other such information storage, transmission, or display devices.

(18) The foregoing has outlined rather broadly certain features and technical advantages of embodiments of the present invention in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter that form the subject of the claims of the invention. It should be appreciated by those having ordinary skill in the

art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same or similar purposes. It should also be realized by those having ordinary skill in the art that such equivalent constructions do not depart from the spirit and scope of the invention as set forth in the appended claims. Additional features will be better understood from the following description when considered in connection with the accompanying figures. It is to be expressly understood, however, that each of the figures is provided for the purpose of illustration and description only and is not intended to limit the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the disclosed system and methods, reference is now made to the following descriptions taken in conjunction with the accompanying drawings.
- (2) FIG. 1 is a block diagram of a system according to some embodiments of the disclosure.
- (3) FIG. 2 is a flow diagram of an example transaction employing a method for disclosing minimally required personal information of a user to a service provider according to some embodiments of the disclosure.
- (4) FIG. 3 is a flow chart of a method for disclosing minimally required personal information of a user to a service provider according to some embodiments of the disclosure.
- (5) FIG. 4 is a schematic block diagram of an example information handling system according to some embodiments of the disclosure.

DETAILED DESCRIPTION

- (6) FIG. 1 illustrates an example computer network **10** (e.g., a telecommunications network) that may be used to implement various aspects of the present application. Generally, the computer network **10** includes various devices of a system communicating and functioning together in the gathering, transmitting, and/or requesting of data to complete one or more services in a manner that safeguards against linking personal information of a user to a public identity of the user and that provides the user with control over what information is shared with various service providers.
- (7) As illustrated, a communications network **106** allows for communication in the computer network **10**. The communications network **106** may include one or more wireless networks such as, but not limited to one or more of a Local Area Network (LAN), Wireless Local Area Network (WLAN), a Personal Area Network (PAN), Campus Area Network (CAN), a Metropolitan Area Network (MAN), a Wide Area Network (WAN), a Wireless Wide Area Network (WWAN), Global System for Mobile Communications (GSM), Personal Communications Service (PCS), Digital Advanced Mobile Phone Service (D-Amps), Bluetooth, Wi-Fi, Fixed Wireless Data, 2G, 2.5G, 3G, 4G, LTE networks, enhanced data rates for GSM evolution (EDGE), General packet radio service (GPRS), enhanced GPRS, messaging protocols such as, TCP/IP, SMS, MMS, extensible messaging and presence protocol (XMPP), real time messaging protocol (RTMP), instant messaging and presence protocol (IMPP), instant messaging, USSD, IRC, or any other wireless data networks or messaging protocols. The communications network **106** may also include wired networks.
- (8) In the computer network **10**, an information handling system **100** communicates with a first computing device **140** and a second computing device **150** over the communications network **106**. For example, the information handling system **100** may be associated with a user, the first computing device **140** may be associated with an online service provider, and the second computing device **150** may be associated with a secondary service provider. In various aspects, the information handling system **100** may be in communication with a single computing device or more than two computing devices. For example, the description of FIG. 2 references computing devices **160**, **170**. Each of the computing devices **140**, **150**, **160**, and **170** may be or include an

information handling system.

(9) The information handling system **100** includes a memory **102** in communication with a processor **104**. The memory **102** stores personally identifying information **110** associated with the user. The personally identifying information **110** includes any digital data that relates to the user. For example, the personally identifying information **110** may include some or all of an identifier (e.g., a globally unique identifier (GUID) or a universally unique identifier (UUID)), biometric information, demographic information, medical information, contact information, payment information, address information, social media activity, user preferences (e.g., music listened to, books read, news filters), internet search history, or the like. The personally identifying information **110** also includes digital data that includes information that relates to the user. For example, an invoice for goods or services may include an identification of the goods or services that the user is purchasing or has purchased, a brand of the goods or services, clothing size information, color selection, or the like, all of which is information related to the user.

(10) A single interaction (e.g., transaction) with a digital service provider may involve a fulfillment chain of services performed by multiple other secondary services providers to complete. For example, a transaction to purchase a clothing article from an online retailer involves a separate delivery service provider delivering the clothing article to the address of the purchaser. As such, different service providers often need different pieces of the personally identifying information **110** to complete their respective service. The information handling system **100** can transmit various portions **112A**, **112B** of the personally identifying information **110** depending on the service provider being sent the information. For example, the portion **112A** may be transmitted to the first computing device **140** associated with the online service provider, as the portion **112A** is all that the online service provider needs to perform the service of the online service provider, and the portion **112B** may be transmitted to the second computing device **150** associated with the second service provider, as the portion **112B** is all that the second service provider needs to perform the service of the second service provider. In an example, the portion **112A** may include biometric information of the user whereas the portion **112B** may include payment information of the user. In another example, the portion **112A** may include address information, and contact information, and medical information of the user whereas the portion **112B** may include payment information of the user. In addition to the portions **112A**, **112B**, a remaining portion of the personally identifying information **110** includes the rest of the personally identifying information **110** that is not included in portions **112A**, **112B**. Any suitable quantity of portions of the personally identifying information **110** may be transmitted by the information handling system **100** and each portion may include any suitable quantity of information.

(11) The information handling system **100** may also employ security measures that reduce the likelihood that any other party gains access to the portion **112A** or **112B** of the personally identifying information **110** other than the service provider being sent the portion **112A** or **112B**. For example, the information handling system **100**, in a secure area of the memory **102**, may create a one-time use cryptographic key pair including a private key **122A** and a public key **124A** for encryption and decryption purposes during the entirety of the single interaction (e.g., transaction) with the online service provider. The information handling system **100** initiates a transaction or other interaction with the digital service provider by encrypting (e.g., cryptographically signing) the public key **124A** with a public key **124B** associated with the online service provider. The public key **124B** may be publicly accessible to the information handling system **100** or the first computing device **140** may transmit the public key **124B** to the information handling system **100** in response to receiving an indication to do so from the information handling system **100**. The public key **124B** is part of a key pair created by the first computing device **140** that includes a private key **122B** and the public key **124B**. In some aspects, the key pair created by the first computing device **140** is a one-time use key pair for cryptographic signing purposes only during the entirety of the single interaction with the information handling system **100**, similar to the key pair created by the

information handling system **100**. The first computing device **140** can decrypt the encrypted public key **124A**, in a secure area of memory of the first computing device **140**, using the private key **122B**. When transmitting information to the information handling system **100**, the first computing device **140** can encrypt that information with the public key **124A** of the information handling system **100**, and upon receipt the information handling system **100** can then decrypt that encrypted information with the private key **122A**.

(12) The first computing device **140** may share the public key **124A** with the computing devices of other service providers that perform secondary services, such as with the second computing device **150**. When the secondary service provider requires a portion of the personally identifying information **110** to complete the respective service, the second computing device **150** can encrypt a request with the public key **124A** and transmit the encrypted request to the information handling system **110**. The information handling system **110** can then decrypt the encrypted request, in a secure area of the memory **102**, with the private key **122A**. Based on the request, the information handling system **100** can encrypt the portion **112A**, **112B** of the personally identifying information **110** that is requested with a public key **124C** associated with the second computing device **150**. The public key **124C** may be publicly accessible to the information handling system **100** or the second computing device **150** may transmit the public key **124C** to the information handling system **100**. The public key **124C** is part of a key pair created by the second computing device **150** that includes a private key **122C** and the public key **124C**. In some aspects, the key pair created by the second computing device **150** is a one-time use key pair for cryptographic signing purposes only during the entirety of the single interaction with the information handling system **100**, similar to the key pair created by the information handling system **100**. The second computing device **150** can decrypt the encrypted portion **112A**, **112B**, in a secure area of memory of the second computing device **150**, using the private key **122C**.

(13) The process of exchanging encrypted requests for information and encrypted portions **112A**, **112B** of the personally identifying information **110** repeats for every service provider in the fulfillment chain that requires particular information of the personally identifying information **110** in order to complete a respective service. When each and every service in the fulfillment chain is complete and the interaction with the digital service provider is finished, the information handling system **100** may discard the private key **122A** and the public key **124A**. Similarly, each of the first computing device **140** and the second computing device **150** may discard their respective key pairs. At a later point in time, when the user wishes to initiate a new interaction with the same digital service provider or a different digital service provider, the information handling system **100** may create a new one-time use cryptographic key pair including a new private key and a new public key for cryptographic signing purposes only during the entirety of the new interaction. Similarly, the computing devices of the service providers involved in the fulfillment chain of the new interaction may create their own respective one-time use cryptographic key pairs.

(14) The memory **102** may further store user policy settings **114**. The user policy settings **114** allow for an automatic data flow between the information handling system **100** and the first computing device **140** and between the information handling system **100** and the second computing device **150**. For instance, the user policy settings **114** include predetermined criteria for how the personally identifying information **110** can be disseminated such that portions of the personally identifying information **110** can be shared or not shared with service providers without requiring further user input. The user policy settings **114** can include, for example, specific service providers for which sharing is allowed, specific service providers for which sharing is prohibited, a reputability threshold that a service provider must meet in order for sharing to be allowed, policies that a service provider must or must not have for sharing to be allowed, and the like.

(15) The information handling system **100** can obtain information about service providers to determine if certain conditions of the user policy settings **114** are met by communicating with a service provider database **130** that stores information on service providers. For example, the service

provider database **130** may store one or both of service provider credentials **132** or service provider history **134**. The service provider credentials **132** may include information about a service provider, such as ratings or scorings of the service provider. The service provider history **134** may include past instances of information security issues with the service provider and the like. Each of the service provider credentials **132** and service provider history **134** may be used to evaluate a reputability of the service provider. For example, the service provider credentials **132** may indicate that a service provider has a business rating below a threshold which results in the service provider having a reputability below the reputability threshold of the user policy settings **114**. As a result, the information handling system **100** does not share any of the personally identifying information **110** with this service provider.

(16) FIG. **2** is a flow diagram of a process **200** that includes the transfer of data between the information handling system **100** associated with a user **202** and computing devices **140**, **150**, **160**, **170** associated with respective service providers. For ease of description, the user **202** can be considered to be purchasing a clothing article from a digital retail service provider **215**. The key symbols in FIG. **2** indicate that a particular block of information is encrypted (e.g., cryptographically signed).

(17) The process **200** involves a fulfillment chain including a series of respective services performed by the digital retail service provider **215**, a transaction service provider **216**, a fulfillment service provider **217**, and a delivery service provider **218**. Stated differently, the delivery service provider **218** performs a delivery service based on the completion of a fulfillment service by the fulfillment service provider **217**, the fulfillment service provider **217** performs the fulfillment service based on the completion of a transaction service by the transaction service provider **216**, and the transaction service provider **216** performs the transaction service based on the completion of a retail service performed by the digital retail service provider **215**. Throughout the process **200**, each respective service provider is transmitted only the portion of the personally identifying information **110** of the user **202** that each respective service provider needs to perform each respective service.

(18) The process **200** may be initiated by the user **202** indicating to the information handling system **100**, via a user interface, that the user **202** wishes to complete a transaction with the digital retail service provider **215** to purchase a clothing article. In response, the information handling system **100** creates the one-time use key pair including the private key **122A** and the public key **124A**. The information handling system **100** then initiates a transaction with the digital retail service provider **215** by encrypting the public key **124A** using the public key **124B** initially created by the computing device **140** associated with the digital retail service provider **215**, and transmitting the encrypted public key **124A** to the computing device **140**. The computing device **140** decrypts the encrypted public key **124A** using the private key **122B** in a secure area of the network of the digital retail service provider **215**.

(19) The computing device **140** also creates a transaction manifest **204** that includes a transaction identifier **206** and appended transaction information. The transaction identifier **206** may be, for example, a globally unique identifier (GUID) or a universally unique identifier (UUID). The transaction information may include, for example, terms of the transaction, a list and description of the items to be fulfilled or actioned, and a list of authorized service providers in the fulfillment chain and their respective public keys. For example, the transaction manifest **204** includes terms and conditions (e.g., price) for purchasing the clothing article, a description of the clothing article to be purchased (e.g., size, color etc.), and a list of the transaction service provider **216**, the fulfillment service provider **217**, and the delivery service provider **218** along with their respective public keys. In some aspects, the transaction information may include a sequence of operations of the transaction. The transaction manifest **204**, however, does not contain any of the personally identifiable information **110** of the user **202**, such as the name, address, or payment information of the user **202**, because none of the personally identifiable information **110** has been provided to the

digital retail service provider **215**.

(20) The computing device **140** of the digital retail service provider **215** then encrypts the transaction manifest **204** using the public key **124A** associated with the information handling system **100** and transmits the encrypted transaction manifest **204** to the information handling system **100**. The information handling system **100** decrypts the encrypted transaction manifest **204** using the public key **124B** associated with the computing device **140**. At this point, the user **202** is prompted to approve, deny, or modify the transaction manifest **204**, the answer to which is transmitted in a message **205** to the computing device **140** by the information handling system **100**. If the message **205** indicates that the transaction manifest **204** is denied, then the private key **122A** and the public key **124A** are each discarded. The key pairs of each respective service provider may also be discarded by the respective computing devices. If the message **205** indicates a modification to the transaction manifest **204**, the transaction manifest **204** will loop between the information handling system **100** and the computing device **140** until the transaction manifest **204** is approved or denied.

(21) Proceeding as if the message **205** indicates approval of the transaction manifest **204**, the computing device **140** thereafter encrypts at least the transaction identifier **206** and the public key **124A** using the public key **124C** initially created by the computing device **150** associated with the transaction service provider **216**, and transmits the encrypted transaction identifier **206** and public key **124A** to the computing device **150**.

(22) Subsequent to the message **205** indicating approval of the transaction manifest **204**, the computing device **140** may also transmit the transaction identifier **206**, the public key **124A**, and triggering conditions **210A**, **210B** to each of the remaining service providers in the fulfillment chain of the process **200**. The triggering conditions **210A**, **210B** indicate to a computing device of a respective service provider when to initiate and terminate the service of the respective service provider. For example, the computing device **104** may encrypt each of the transaction identifier **206**, the public key **124A**, and triggering conditions **210A** using a public key **124D** initially created by a computing device **160** associated with the fulfillment service provider **217**, and transmit the encrypted transaction identifier **206**, the public key **124A**, and triggering conditions **210A** to the computing device **160**. In this example, the triggering conditions **210A** may indicate for the computing device **160** to initiate the fulfillment service in response to receiving a trigger **206A** from the computing device **150** that indicates the payment is complete, and to terminate the fulfillment service upon transmitting a trigger **206B** to a computing device **170** associated with the delivery service provider **118**. The trigger **206B** may indicate that the clothing article has been successfully packaged and is awaiting pickup.

(23) The computing device **104** may similarly encrypt each of the transaction identifier **206**, the public key **124A**, and triggering conditions **210B** using a public key **124E** initially created by the computing device **170** associated with the delivery service provider **218**, and transmit the encrypted transaction identifier **206**, the public key **124A**, and triggering conditions **210B** to the computing device **160**. In this example, the triggering conditions **210B** may indicate for the computing device **170** to initiate the delivery service in response to receiving the trigger **206B** from the computing device **160**, and to terminate the delivery service upon transmitting a trigger **206C** to the computing device **140**. The trigger **206C** may indicate that the packaged clothing article has been successfully picked up and is in route to the user **202**.

(24) Returning to the computing device **150**, in some aspects, the computing device **150** is able to identify a payment due for the clothing article based on the transaction identifier **206**. In other aspects, the computing device **140** transmits the payment due, encrypted using the public key **124C**, to the computing device **150**. The computing device **150** decrypts the encrypted transaction identifier **206** and public key **124A** using the private key **122C**. The computing device **150** then needs payment information from the user **202** in order to perform the transaction service of anonymously brokering payment between a private banking service of the user **202** and a

transaction processor of the digital retail service provider **215** by processing the payment. As such, the computing device **150** encrypts a request **207A** for payment information using the public key **124A** and transmits the encrypted request **207A** to the information handling system **100**.

(25) The information handling system **100** decrypts the encrypted request **207A** to determine that the computing device **150** requests payment information. In some aspects, the user **202** may manually indicate whether the transaction service provider **216** associated with the computing device **150** is an authorized party to which payment information **209** (e.g., portion **112A**) of the personally identifying information **110** may be sent. In other aspects, the information handling system **100** may automatically determine whether the transaction service provider **216** is an authorized party based on the user policy settings **114**. If the user **202** or the user policy settings **114** prohibit releasing the payment information **209** to the transaction service provider **216**, then the transaction is terminated the respective key pairs are discarded. Proceeding as if the user **202** of the user policy settings **114** indicate that the payment information **209** can be released to the transaction service provider **216**, the information handling system **100** encrypts the payment information **209** using the public key **124C** and transmits the encrypted payment information **209** to the computing device **150**. The payment information **209** may include credit card information, bank routing details, and the like.

(26) The computing device **150** decrypts the encrypted payment information **209** using the private key **122C** and thereafter processes the payment based on the payment information **209**. In this way, the transaction service provider **216** does not gain any knowledge of the public identity of the user **202**, the clothing article that has been purchased, or where the clothing article is going. Subsequent to the payment being processed, the transaction service provider **216** transmits the trigger **206A** to the computing device **160** associated with the fulfillment service provider **217**.

(27) The fulfillment service provider **217** does not need any of the personally identifying information **110** of the user **202** in order to complete the fulfillment service, and therefore none is requested. Instead, the computing device **160** generates a notification in response to receiving the trigger **206A** for the fulfillment service provider **217** to initiate the fulfillment service. The computing device **160** may also decrypt the encrypted transaction identifier **206** using a private key **122D** of the key pair initially created by the computing device **160** that includes the public key **124D**. The fulfillment service provider **217** thereafter packages the clothing article ordered by the user **202** based on transaction identifier **206**. Once the fulfillment service provider **217** (e.g., a warehouse worker) indicates that the clothing article is successfully packaged, the computing device **160** transmits the trigger **206B** to the computing device **170** associated with the delivery service provider **218**. The fulfillment service provider **217** may affix a label including the transaction identifier **206** to the package. In this way, the fulfillment service provider **217** does not gain any knowledge of a public identity of who the package is for or where the package is going.

(28) The delivery service provider **218** requires address information indicating where the user **202** wants the packaged clothing article to be sent. As such, in response to receiving the trigger **206B**, the computing device **170** encrypts a request **207B** for address information using the public key **124A**, and transmits the encrypted request **207B** to the information handling system **100**.

(29) The information handling system **100** decrypts the encrypted request **207B** to determine that the computing device **170** requests address information. In some aspects, the user **202** may manually indicate whether the delivery service provider **218** associated with the computing device **170** is an authorized party to which address information **208** (e.g., second portion **112B**) of the personally identifying information **110** may be sent. In other aspects, the information handling system **100** may automatically determine whether the delivery service provider **218** is an authorized party based on the user policy settings **114**. If the user **202** or the user policy settings **114** prohibit releasing the address information **208** to the delivery service provider **218**, then the transaction may terminated and the respective key pairs discarded, in some aspects. In such aspects, the user **202** may be refunded payment. In other aspects, a new delivery service provider that is

authorized may be determined. Proceeding as if the user **202** of the user policy settings **114** indicate that the address information **208** can be released to the delivery service provider **218**, the information handling system **100** encrypts the address information **208** using the public key **124E** and transmits the encrypted address information **208** to the computing device **170**. The address information **208** may include details of the address where the user **202** wants the package delivered. (30) Subsequent to the receipt of the encrypted address information **208**, the computing device **170** decrypts the encrypted address and in **208** using a private key **124E** of the key pair initially created by the computing device **170** that includes the public key **124E**. Based on the address information **208**, the delivery service provider **218** is able to complete the delivery service by delivering the package to the address indicated. The delivery service provider **218** may affix a label containing the address to the package to assist in routing and delivery, but the delivery service provider **218** does not gain any knowledge of the package contents other than what is required to safely deliver the package.

(31) In some aspects, the computing device **170** may transmit a trigger **206C** to the computing device **140** that indicates the package is in route to the user **202**. Based on receiving the trigger **206C**, the computing device **140** may transmit a message **212** to the information handling system **100**. For example, the message **212** may be a shipping confirmation email. In some aspects, the computing device **140** may encrypt the message **212** using the public key **124A** prior to transmittal. In some aspects, each of the triggers **206A**, **206B**, **206C** may be encrypted prior to transmittal to a the respective recipient.

(32) In some aspects, rather than each of the computing devices **140**, **150**, **160**, **170** of the service providers communicating directly with the information handling system **100**, one or more of the computing devices **140**, **150**, **160**, **170** may communicate with a computing device associated with a brokering service provider authorized to transmit the information on the behalf of the user **202**.

(33) In this way, no individual service provider in the fulfillment chain of the process **200** has sufficient information to compromise the private information of the commerce originator or recipient, which in this case are both the user **202**, though the originator or recipient could be a different individual than the user **202**. For example, in some aspects, the user **202** may purchase the clothing article **220** for a friend, rather than for the user **202**, using the techniques described herein, even if the user **202** does not know certain information about the friend such as the clothing size or address of the friend. For example, the user **202** may specify an anonymous digital identity of the friend (e.g., an identity token that anonymously identifies the friend) as the recipient. When information regarding the friend is required, a respective service provider requests that information from an information handling system associated with the friend. The information handling system associated with the friend releases or withholds the requested information depending on the policy settings associated with the friend. If the policy settings of the friend prohibit transmittal of required data to a particular service provider in the process, then the transaction cannot be completed.

(34) It will be appreciated that the technologically-based information security techniques described herein are applicable to the delivery of digital goods or services as well as physical goods (e.g., the clothing article). For example, in the case of a digital good or service, such as a digital workout program offered by a digital service provider, subsequent to the transaction service provider **216** performing the transaction service, the user **202** is granted access to the digital workout program. For example, a license, file, binary, or other digital information associated with the digital workout program is linked to the identifier **205** of the user **202** for a term specified in the transaction settlement. The user **202** can then access the digital workout program, subject to the terms and conditions of the digital service provider, by authenticating against the identifier **205** of the user **202**.

(35) In another example of digital goods and services, the user **202** may interact with a news website to tailor the delivery of news to the user **202**. For instance, the user **202** has a set of specific

interests and preferences concerning news and cultural happenings and the user **202** desires content presented to the user **202** by the news website be curated and filtered to reflect these interests and preferences. As such, out of the personally identifying information **110**, these specific interests and preferences may be encrypted and transmitted to the news website, but the news website has no knowledge of any other information that is linked to a public identity of the user **202**. In some aspects, the specific interests and preferences of the user **202** may apply generically to any digital platform that presents content to the user **202**. In these aspects, the specific interests and preferences can be transmitted to any content-providing digital platform that the user **202** interacts with, rather than the user **202** having to input settings for each individual platform that each have different settings to choose from. In this way, the technologically-based information security techniques enable a more consistent experience for the user **202** across various content-providing digital platforms. The techniques further reduce the market for selling the collected interests and preferences of the user by service providers because the interests and preferences are not tied to any other information of the user, such as demographic information, and therefore the collected interests and preferences have reduced utility.

(36) In another example, when browsing various clothing articles on a website of the digital retail service provider **215**, the user **202** may desire to see a virtual rendering of a particular clothing item on a form approximating a body type of the user **202** to determine whether the clothing article suits the preferences of the user **202**. A rendering service provider needs only biometric information of the user **202** to perform the rendering service of generating the virtual rendering. As such, out of the personally identifying information **110** of the user **202**, only biometric information of the user **202** is encrypted and transmitted to the rendering service provider. In an example, the biometric information may include disidentified body type weights that the rendering service provider uses to return a virtual rendering of the clothing article on a generic body type. In this example, once the virtual rendering is received by the information handling system **100** of the user **202**, the information handling system **100** may composite identifiable body elements of the user **202** into the virtual rendering. In this way, the rendering service provider has no knowledge of the public identity of the user or any other information related to the purchase of the clothing article.

(37) FIG. 3 is a flow chart of a method **400** for disclosing minimally required personal information of a user (e.g., user **202**) to a service provider. The method **400** includes, at block **402**, receiving a first encrypted request (e.g., request **207A**) for a first portion (e.g., portion **112A**) of the personally identifying information (e.g., personally identifying information **110**) associated with the user **202**. The encrypted request **207A** is encrypted using a first public key (e.g., public key **124C**) of a first key pair including a first private key (e.g., private key **122C**) and the first public key **124C**.

(38) A completion of a first service (e.g., transaction service) by first service provider (e.g., the transaction service provider **216**) is based on the first portion **112A**. For example, the transaction service provider **216** requires the first portion **112A** (e.g., payment information **209**) of the personally identifying information **110** in order to complete the transaction service. Otherwise, the transaction service provider **216** cannot provide the transaction service. In some aspects, of the personally identifying information **110** associated with the user **202**, the completion of the first service requires only the first portion **112A**. For example, the transaction service provider **206** does not need the biometric information of the user or any other personally identifying information **110** other than the payment information **209** to perform the transaction service.

(39) In the method **400**, the first service is part of a process (e.g., process **200**) that includes the first service performed by the transaction service provider **216** and a second service (e.g., delivery service) performed by a second service provider (e.g., the delivery service provider **218**) different than the first service provider. The delivery service is different than the transaction service, and the delivery service is performed by the delivery service provider **218** based on a completion of the transaction service by the transaction service provider **218**. For example, in order for the delivery service provider **218** to deliver the package, the fulfillment service provider **217** must package the

item, and in order for the fulfillment service provider **217** to package the item, the transaction service provider **216** must complete brokering the settlement.

(40) In various aspects, the personally identifying information **110** associated with the user includes biometric information, demographic information, contact information, user preferences, payment information (e.g., payment information **209**), or address information (e.g., address information **208**), or a combination thereof.

(41) At block **404**, the first encrypted request **207A** is decrypted using the first private key **122A**.

(42) At block **406**, the payment information **209** of the personally identifying information **110** is encrypted using a second public key (e.g., public key **124C**) associated with the transaction service provider **216**.

(43) At block **408**, the encrypted payment information **209** is transmitted to a first computing device (e.g., computing device **150**) associated with the transaction service provider **216**. In at least some aspects, the personally identifying information **110** associated with the user **202** consists of the payment information **209**, the address information **208**, and a remaining portion, and the address information **208** and the remaining portion are not transmitted to the first computing device **150**. For example, neither the address information **208** nor any personally identifying information **110** other than the payment information **209** is transmitted to the first computing device **150** of the transaction service provider **216**.

(44) At block **410**, subsequent to a completion of the transaction service by the transaction service provider **216**, a second encrypted request (e.g., request **207B**) for a second portion (e.g., portion **112B**) of the personally identifying information **110** associated with the user **202** is received. The second encrypted request **207B** is encrypted using the first public key **124A**. In this example, the portion **112B** is the address information **208** associated with the user **202**, which is different than the payment information **209**. A completion of the delivery service is based on the payment information **209**. For example, the delivery service provider **218** needs the address information **208** of the personally identifying information **110** in order to complete the delivery service. Otherwise, the delivery service provider **218** cannot provide the delivery service. In some aspects, of the personally identifying information **110** associated with the user **202**, the completion of the delivery service requires only the address information **208**. For example, in such aspects, the delivery service provider **218** does not need the payment information **209** of the user or any other personally identifying information **110** other than the address information **208** to perform the delivery service. In some aspects, the delivery service provider **218** may require some information about the contents of the package in order to safely complete the delivery service.

(45) At block **412**, the second encrypted request **207B** is decrypted using the first private key **122A**.

(46) At block **414**, the address information **208** of the personally identifying information **110** is encrypted using a third public key (e.g., public key **124E**) associated with the delivery service provider **218**.

(47) At block **416**, the encrypted address information **208** is transmitted to a second computing device (e.g., the computing device **170**) associated with the delivery service provider **218**. In at least some aspects, the personally identifying information **110** associated with the user **202** consists of the payment information **209**, the address information **208**, and a remaining portion, and the payment information **209** and the remaining portion are not transmitted to the second computing device **170**. For example, neither the payment information **209** nor any personally identifying information **110** other than the address information **208** is transmitted to the second computing device **170** of the delivery service provider **218**.

(48) In some aspects, the method **400** may further include discarding the first private key **122A** and the first public key **124A** subsequent to a completion of the process **200**.

(49) These example embodiments describe and illustrate various computer-based techniques that improve upon the technical field of digital information security as a whole. The techniques further improve upon an information handling system by enabling the information handling system to

serve as a technological tool that individuals can use to have full technological control over the dissemination of their own digital personally identifying information via a cryptographic signing mechanism specific to a particular individual. With this technological control and predetermined criteria set by a user, the improved information handling system transmits the minimally required personal information of a user to a computing device associated with a service provider for a particular purpose without a link to a public identity of the user.

(50) For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components. One example configuration of an information handling system is described with reference to FIG. 4.

(51) FIG. 4 illustrates an example information handling system **500**. Information handling system **500** may include a processor **502** (e.g., a central processing unit (CPU)), a memory (e.g., a dynamic random-access memory (DRAM)) **504**, and a chipset **506**. In some embodiments, one or more of the processor **502**, the memory **504**, and the chipset **506** may be included on a motherboard (also referred to as a mainboard), which is a printed circuit board (PCB) with embedded conductors organized as transmission lines between the processor **502**, the memory **504**, the chipset **506**, and/or other components of the information handling system. The components may be coupled to the motherboard through packaging connections such as a pin grid array (PGA), ball grid array (BGA), land grid array (LGA), surface-mount technology, and/or through-hole technology. In some embodiments, one or more of the processor **502**, the memory **504**, the chipset **506**, and/or other components may be organized as a System on Chip (SoC).

(52) The processor **502** may execute program code by accessing instructions loaded into memory **504** from a storage device, executing the instructions to operate on data also loaded into memory **504** from a storage device, and generate output data that is stored back into memory **504** or sent to another component. The processor **502** may include processing cores capable of implementing any of a variety of instruction set architectures (ISAs), such as the x86, POWERPC®, ARM®, SPARC®, or MIPS® ISAs, or any other suitable ISA. In multi-processor systems, each of the processors **502** may commonly, but not necessarily, implement the same ISA. In some embodiments, multiple processors may each have different configurations such as when multiple processors are present in a big-little hybrid configuration with some high-performance processing cores and some high-efficiency processing cores. The chipset **506** may facilitate the transfer of data between the processor **502**, the memory **504**, and other components. In some embodiments, chipset **506** may include two or more integrated circuits (ICs), such as a northbridge controller coupled to the processor **502**, the memory **504**, and a southbridge controller, with the southbridge controller coupled to the other components such as USB **510**, SATA **520**, and PCIe buses **508**. The chipset **506** may couple to other components through one or more PCIe buses **508**.

(53) Some components may be coupled to one bus line of the PCIe buses **508**, whereas some

components may be coupled to more than one bus line of the PCIe buses **508**. One example component is a universal serial bus (USB) controller **510**, which interfaces the chipset **506** to a USB bus **512**. A USB bus **512** may couple input/output components such as a keyboard **514** and a mouse **516**, but also other components such as USB flash drives, or another information handling system. Another example component is a SATA bus controller **520**, which couples the chipset **506** to a SATA bus **522**. The SATA bus **522** may facilitate efficient transfer of data between the chipset **506** and components coupled to the chipset **506** and a storage device **524** (e.g., a hard disk drive (HDD) or solid-state disk drive (SDD)) and/or a compact disc read-only memory (CD-ROM) **526**. The PCIe bus **508** may also couple the chipset **506** directly to a storage device **528** (e.g., a solid-state disk drive (SDD)). A further example of an example component is a graphics device **530** (e.g., a graphics processing unit (GPU)) for generating output to a display device **532**, a network interface controller (NIC) **540**, and/or a wireless interface **550** (e.g., a wireless local area network (WLAN) or wireless wide area network (WWAN) device) such as a Wi-Fi® network interface, a Bluetooth® network interface, a GSM® network interface, a 3G network interface, a 4G LTE® network interface, and/or a 5G NR network interface (including sub-6 GHz and/or mmWave interfaces).

(54) The chipset **506** may also be coupled to a serial peripheral interface (SPI) and/or Inter-Integrated Circuit (I2C) bus **560**, which couples the chipset **506** to system management components. For example, a non-volatile random-access memory (NVRAM) **570** for storing firmware **572** may be coupled to the bus **560**. As another example, a controller, such as a baseboard management controller (BMC) **580**, may be coupled to the chipset **506** through the bus **560**. BMC **580** may be referred to as a service processor or embedded controller (EC). Capabilities and functions provided by BMC **580** may vary considerably based on the type of information handling system. For example, the term baseboard management system may be used to describe an embedded processor included at a server, while an embedded controller may be found in a consumer-level device. As disclosed herein, BMC **580** represents a processing device different from processor **502**, which provides various management functions for information handling system **500**. For example, an embedded controller may be responsible for power management, cooling management, and the like. An embedded controller included at a data storage system may be referred to as a storage enclosure processor or a chassis processor.

(55) System **500** may include additional processors that are configured to provide localized or specific control functions, such as a battery management controller. Bus **560** can include one or more busses, including a Serial Peripheral Interface (SPI) bus, an Inter-Integrated Circuit (I2C) bus, a system management bus (SMBUS), a power management bus (PMBUS), or the like. BMC **580** may be configured to provide out-of-band access to devices at information handling system **500**. Out-of-band access in the context of the bus **560** may refer to operations performed prior to execution of firmware **572** by processor **502** to initialize operation of system **500**.

(56) Firmware **572** may include instructions executable by processor **102** to initialize and test the hardware components of system **500**. For example, the instructions may cause the processor **502** to execute a power-on self-test (POST). The instructions may further cause the processor **502** to load a boot loader or an operating system (OS) from a mass storage device. Firmware **572** additionally may provide an abstraction layer for the hardware, such as a consistent way for application programs and operating systems to interact with the keyboard, display, and other input/output devices. When power is first applied to information handling system **500**, the system may begin a sequence of initialization procedures, such as a boot procedure or a secure boot procedure. During the initialization sequence, also referred to as a boot sequence, components of system **500** may be configured and enabled for operation and device drivers may be installed. Device drivers may provide an interface through which other components of the system **500** can communicate with a corresponding device. The firmware **572** may include a basic input-output system (BIOS) and/or include a unified extensible firmware interface (UEFI). Firmware **572** may also include one or

more firmware modules of the information handling system. Additionally, configuration settings for the firmware **572** and firmware of the information handling system **500** may be stored in the NVRAM **570**. NVRAM **570** may, for example, be a non-volatile firmware memory of the information handling system **500** and may store a firmware memory map namespace **500** of the information handling system. NVRAM **570** may further store one or more container-specific firmware memory map namespaces for one or more containers concurrently executed by the information handling system.

(57) Information handling system **500** may include additional components and additional busses, not shown for clarity. For example, system **500** may include multiple processor cores (either within processor **502** or separately coupled to the chipset **506** or through the PCIe buses **508**), audio devices (such as may be coupled to the chipset **506** through one of the PCIe buses **508**), or the like. While a particular arrangement of bus technologies and interconnections is illustrated for the purpose of example, one of skill will appreciate that the techniques disclosed herein are applicable to other system architectures. System **500** may include multiple processors and/or redundant bus controllers. In some embodiments, one or more components may be integrated together in an integrated circuit (IC), which is circuitry built on a common substrate. For example, portions of chipset **506** can be integrated within processor **502**. Additional components of information handling system **500** may include one or more storage devices that may store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display.

(58) In some embodiments, processor **502** may include multiple processors, such as multiple processing cores for parallel processing by the information handling system **500**. For example, the information handling system **500** may include a server comprising multiple processors for parallel processing. In some embodiments, the information handling system **500** may support virtual machine (VM) operation, with multiple virtualized instances of one or more operating systems executed in parallel by the information handling system **500**. For example, resources, such as processors or processing cores of the information handling system may be assigned to multiple containerized instances of one or more operating systems of the information handling system **500** executed in parallel. A container may, for example, be a virtual machine executed by the information handling system **500** for execution of an instance of an operating system by the information handling system **500**. Thus, for example, multiple users may remotely connect to the information handling system **500**, such as in a cloud computing configuration, to utilize resources of the information handling system **500**, such as memory, processors, and other hardware, firmware, and software capabilities of the information handling system **500**. Parallel execution of multiple containers by the information handling system **500** may allow the information handling system **500** to execute tasks for multiple users in parallel secure virtual environments.

(59) The flow diagram of FIG. **2** or flow chart of FIG. **3** are generally set forth as a logical flow chart diagram. As such, the depicted order and labeled steps are indicative of aspects of the disclosed method. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more steps, or portions thereof, of the illustrated method. Additionally, the format and symbols employed are provided to explain the logical steps of the method and are understood not to limit the scope of the method. Although various arrow types and line types may be employed in the flow chart diagram, they are understood not to limit the scope of the corresponding method. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the method. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted method. Additionally, the order in which a particular method occurs may or may not strictly adhere to the order of the corresponding steps shown.

(60) If implemented in firmware and/or software, functions described above may be stored as one or more instructions or code on a computer-readable medium. Examples include non-transitory

computer-readable media encoded with a data structure and computer-readable media encoded with a computer program. Computer-readable media includes physical computer storage media. A storage medium may be any available medium that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise random access memory (RAM), read-only memory (ROM), electrically-erasable programmable read-only memory (EEPROM), compact disc read-only memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. Disk and disc includes compact discs (CD), laser discs, optical discs, digital versatile discs (DVD), floppy disks and Blu-ray discs. Generally, disks reproduce data magnetically, and discs reproduce data optically. Combinations of the above should also be included within the scope of computer-readable media.

(61) In addition to storage on computer readable medium, instructions and/or data may be provided as signals on transmission media included in a communication apparatus. For example, a communication apparatus may include a transceiver having signals indicative of instructions and data. The instructions and data are configured to cause one or more processors to implement the functions outlined in the claims.

(62) Although the present disclosure and certain representative advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. Further, a device or system that is configured in a certain way is configured in at least that way, but it can also be configured in other ways than those specifically described. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. For example, although processors are described throughout the detailed description, aspects of the invention may be applied to the design of or implemented on different kinds of processors, such as graphics processing units (GPUs), central processing units (CPUs), and digital signal processors (DSPs). As another example, although processing of certain kinds of data may be described in example embodiments, other kinds or types of data may be processed through the methods and devices described above. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. An information handling system, comprising: a memory storing a private key and personally identifying information associated with a user; and a processor coupled to the memory, wherein the processor is configured to perform operations comprising: receiving a first encrypted request for a first portion of the personally identifying information associated with the user, wherein: the encrypted request is encrypted using a first public key of a first key pair including the private key and the first public key, a completion of a first service by a first service provider is based on the first portion, the first service is part of a process that includes the first service performed by the first service provider and a second service performed by a second service provider different than the first service provider, the second service is different than the first service, and the second service is performed by the second service provider based on a completion of the first service by the first service provider; decrypting the first encrypted request using the private key; encrypting the first portion of the personally identifying information using a second public key associated with the first

service provider; transmitting the first portion that is encrypted to a first computing device associated with the first service provider; receiving, subsequent to a completion of the first service by the first service provider, a second encrypted request for a second portion of the personally identifying information associated with the user, wherein: the second encrypted request is encrypted using the first public key, the second portion is different than the first portion, and a completion of the second service by the second service provider is based on the second portion; decrypting the second encrypted request using the private key; encrypting the second portion of the personally identifying information using a third public key associated with the second service provider; and transmitting the second portion that is encrypted to a second computing device associated with the second service provider.

2. The information handling system of claim 1, wherein the personally identifying information associated with the user consists of the first portion, the second portion, and a remaining portion, and wherein the second portion and the remaining portion are not transmitted to the first computing device.

3. The information handling system of claim 2, wherein, of the personally identifying information associated with the user, the completion of the first service requires only the first portion.

4. The information handling system of claim 3, wherein the first portion and the remaining portion are not transmitted to the second computing device.

5. The information handling system of claim 4, wherein, of the personally identifying information associated with the user, the completion of the second service requires only the second portion.

6. The information handling system of claim 1, wherein the operations further include discarding the private key and the first public key subsequent to a completion of the process.

7. The information handling system of claim 1, wherein the personally identifying information associated with the user includes biometric information, demographic information, contact information, user preferences, payment information, an invoice for goods or services, or address information, or a combination thereof.

8. A method, comprising: receiving a first encrypted request for a first portion of personally identifying information associated with a user, wherein: the encrypted request is encrypted using a first public key of a first key pair including a private key and the first public key, a completion of a first service by a first service provider is based on the first portion, the first service is part of a process that includes the first service performed by the first service provider and a second service performed by a second service provider different than the first service provider, the second service is different than the first service, and the second service is performed by the second service provider based on a completion of the first service by the first service provider; decrypting the first encrypted request using the private key; encrypting the first portion of the personally identifying information using a second public key associated with the first service provider; transmitting the first portion that is encrypted to a first computing device associated with the first service provider; receiving, subsequent to a completion of the first service by the first service provider, a second encrypted request for a second portion of the personally identifying information associated with the user, wherein: the second encrypted request is encrypted using the first public key, the second portion is different than the first portion, and a completion of the second service by the second service provider is based on the second portion; decrypting the second encrypted request using the private key; encrypting the second portion of the personally identifying information using a third public key associated with the second service provider; and transmitting the second portion that is encrypted to a second computing device associated with the second service provider.

9. The method of claim 8, wherein the personally identifying information associated with the user consists of the first portion, the second portion, and a remaining portion, and wherein the second portion and the remaining portion are not transmitted to the first computing device.

10. The method of claim 9, wherein, of the personally identifying information associated with the user, the completion of the first service requires only the first portion.

11. The method of claim 10, wherein the first portion and the remaining portion are not transmitted to the second computing device.
 12. The method of claim 11, wherein, of the personally identifying information associated with the user, the completion of the second service requires only the second portion.
 13. The method of claim 8, further comprising discarding the private key and the first public key subsequent to a completion of the process.
 14. The method of claim 8, wherein the personally identifying information associated with the user includes biometric information, demographic information, contact information, user preferences, payment information, an invoice for goods or services, or address information, or a combination thereof.
 15. A computer program product, comprising: a non-transitory computer readable medium comprising code for performing operations comprising: receiving a first encrypted request for a first portion of personally identifying information associated with a user, wherein: the encrypted request is encrypted using a first public key of a first key pair including a private key and the first public key, a completion of a first service by a first service provider is based on the first portion, the first service is part of a process that includes the first service performed by the first service provider and a second service performed by a second service provider different than the first service provider, the second service is different than the first service, and the second service is performed by the second service provider based on a completion of the first service by the first service provider; decrypting the first encrypted request using the private key; encrypting the first portion of the personally identifying information using a second public key associated with the first service provider; transmitting the first portion that is encrypted to a first computing device associated with the first service provider; receiving, subsequent to a completion of the first service by the first service provider, a second encrypted request for a second portion of the personally identifying information associated with the user, wherein: the second encrypted request is encrypted using the first public key, the second portion is different than the first portion, and a completion of the second service by the second service provider is based on the second portion; decrypting the second encrypted request using the private key; encrypting the second portion of the personally identifying information using a third public key associated with the second service provider; and transmitting the second portion that is encrypted to a second computing device associated with the second service provider.
 16. The computer program product of claim 15, wherein the personally identifying information associated with the user consists of the first portion, the second portion, and a remaining portion, wherein the second portion and the remaining portion are not transmitted to the first computing device, and wherein the first portion and the remaining portion are not transmitted to the second computing device.
 17. The computer program product of claim 15, wherein, of the personally identifying information associated with the user, the completion of the first service requires only the first portion.
 18. The computer program product of claim 17, wherein, of the personally identifying information associated with the user, the completion of the second service requires only the second portion.
 19. The computer program product of claim 15, wherein the operations further include discarding the private key and the first public key subsequent to a completion of the process.
 20. The computer program product of claim 15, wherein the personally identifying information associated with the user includes biometric information, demographic information, contact information, user preferences, payment information, an invoice for goods or services, or address information, or a combination thereof.
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